

9.3 MODBUS Communication Protocol

There are two modes of MODBUS networks communication, ASCII (American Standard Code for information interchange) mode and RTU (Remote Terminal Unit) mode. Users could set the needed communication protocol via parameter P3-02. Apart from these two communication modes, this servo drive also supports function of 03H to access more than one data, 06H to write one character and 10H to write multiple characters. Please refer to the following descriptions.

■ Code Description

~~ASCII Mode:~~

The so-called ASCII mode is using American Standard Code for Information Interchange (ASCII) to transmit the data. Between two stations (Master and Slave) to transmit data 64H, the master will send '6' which represented by 36H of ASCII code and '4' represented by 34H of ASCII code.

ASCII code of digit 0 to 9 and characters A to F is as follows:

Character	'0'	'1'	'2'	'3'	'4'	'5'	'6'	'7'
ASCII code	30H	31H	32H	33H	34H	35H	36H	37H
Character	'8'	'9'	'A'	'B'	'C'	'D'	'E'	'F'
ASCII code	38H	39H	41H	42H	43H	44H	45H	46H

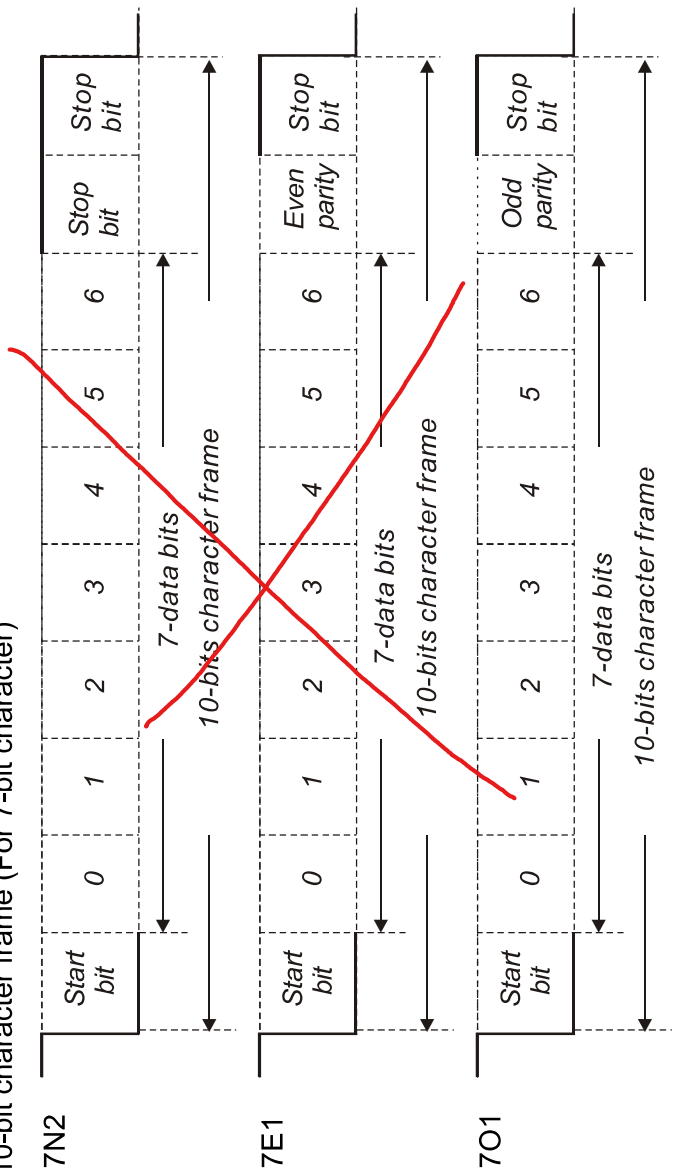
RTU Mode:

Every 8-bit of data is constituted by two 4-bits hexadecimal characters. If data 64H is transmitted between two stations, it will be transmitted directly, which is more efficient than ASCII mode.

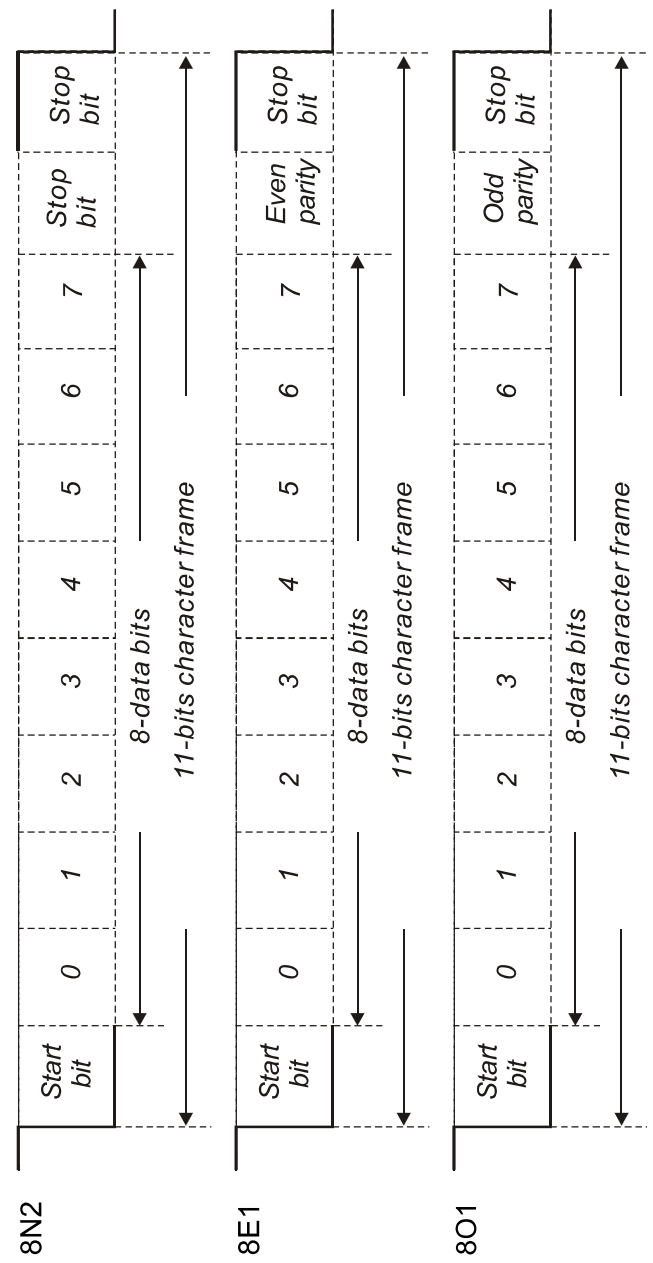
■ Character Structure

Characters will be encoded into the following framing and transmitted in serial. The checking method of different bit is as the following.

10-bit character frame (For 7-bit character)



11-bit character frame (For 8-bit character)



■ Communication Data Structure

~~ASCII Mode:~~

Start	Start character ':' (3AH)
Slave Address	Communication address: 1-byte consists of 2 ASCII codes
Function	Function code: 1-byte consists of 2 ASCII codes
Data (n-1)	Data content: n-word = 2n-byte includes 4n of ASCII code, n<=10
.....	
Data (0)	Error checking: 1-byte consists of 2 ASCII codes
LRC	
End 1	
End 0	

The start character of communication in ASCII mode is colon ':' (ASCII is 3AH), ADR is the ASCII code of two characters. The end code is CR (Carriage Return) and LF (Line Feed). And the communication address, function code, data content, error checking LRC (Longitudinal Redundancy Check), etc are between the start character and end code.

RTU Mode:

Start	A silent interval of more than 10ms
Slave Address	Communication address: 1-byte
Function	Function code: 1-byte
Data (n-1)	Data content : n-word =2n-byte · n<=10
.....	
Data (0)	Error checking: 2-byte
CRC	
End 1	A silent interval of more than 10ms

The start of communication in RTU (Remote Terminal Unit) mode is a silent interval. The end of it is another silent interval. The communication address, function code, data content, error checking CRC (Cyclical Redundancy Check), etc are between the start and the end.

Example 1: function code 03H, access multiple words:

The Master issues the command to the 1st Slave and reads the continuous 2 words starting from the start address 0200H. In response message from the Slave, the content of starting address 0200H is 00B1H and the content of the 2nd data address 0201H is 1F40H. The maximum allowable data in one single access is 10. The calculation of LRC and CRC will be described in next chapter.

ASCII Mode:

Command message (Master):

Start	'.'
Slave Address	'0'
Function	'1'
	'0'
Function	'3'
	'0'
Starting data address	'2'
	'0'
Starting data address	'0'
	'0'
Number of data (In Word)	'0'
	'0'
Number of data (In Word)	'2'
	'F'
LRC Check	'8'
End 1	(0DH)(CR)
End 0	(0AH)(LF)

Response message (Slave):

Start	'.'
Slave Address	'0'
Function	'1'
	'0'
Function	'3'
	'0'
Number of data (In Byte)	'0'
	'4'
Number of data (In Byte)	'0'
	'0'
Content of starting data address 0200H	'B'
	'1'
Content of starting data address 0200H	'1'
	'F'
Content of second data address 0201H	'4'
	'0'
Content of second data address 0201H	'E'
	'8'
LRC Check	
End 1	(0DH)(CR)
End 0	(0AH)(LF)

RTU Mode:

Command message (Master):

Slave Address	01H
Function	03H
Starting data address	02H (High)
	00H (Low)
Number of data (In Word)	00H
	02H
CRC Check Low	C5H (Low)
CRC Check High	B3H (High)

Response message (Slave):

Slave Address	01H
Function	03H
Number of data (In Byte)	04H
Content of starting data address 0200H	00H (High)
	B1H (Low)
Content of second data address 0201H	1FH (High)
	40H (Low)
CRC Check Low	A3H (Low)
CRC Check High	D4H (High)

Please note:

Before and after the transmission in RTU mode, 10ms of silent interval is needed.

Example 2: function code 06H, write single word:

The Master issues command to the 1st Slave and writes data 0064H to address 0200H. The Slave sends the response message to the Master after the writing is completed. The calculation of LRC and CRC will be described in next chapter.

ASCII Mode:

Command message (Master):

Start	','
Slave Address	'0'
Function	'1'
Starting data address	'0'
	'6'
	'0'
	'2'
	'0'
	'0'
	'0'
	'0'
	'6'
	'4'
	'9'
	'3'
LRC Check	
End 1	(0DH)(CR)
End 0	(0AH)(LF)

Response message (Slave):

Start	','
Slave Address	'0'
Function	'1'
Starting data address	'0'
	'6'
	'0'
	'2'
	'0'
	'0'
	'0'
	'0'
	'6'
	'4'
	'9'
	'3'
LRC Check	
End 1	(0DH)(CR)
End 0	(0AH)(LF)

RTU Mode:

Command message (Master):

Address	01H
Slave Function	06H
Starting data address	02H (High)
	00H (Low)
Data content	00H (High)
	64H (Low)
CRC Check Low	89H (Low)
CRC Check High	99H (High)

Response message (Slave):

Address	01H
Slave Function	06H
Starting data address	02H (High)
	00H (Low)
Data content	00H (High)
	64H (Low)
CRC Check Low	89H (Low)
CRC Check High	99H (High)

Please note:

Before and after the transmission in RTU mode, 10ms of silent interval is needed.

Example 3: function code 10H, write multiple words:

The Master issues command to the 1st Slave and writes 0BB8H and 0000H to the starting address 0112H. That is to say, 0112H is written into 0BB8H and 0113H is written into 0000H. The maximum allowable data in one single access is 10. The Slave sends the response message to the Master after the writing is completed. The calculation of LRC and CRC will be described in next chapter.

ASCII Mode:

Command message (Master):

Start	'.'
Slave Address	'0'
	'1'
Function	'1'
	'0'
	'0'
	'1'
Starting data address	'1'
	'1'
	'2'
	'0'
Number of data (In Word)	'0'
	'0'
	'2'
	'0'
Number of data (In Byte)	'0'
	'4'
	'0'
	'B'
The first data content	'B'
	'8'
	'0'
	'0'
The second data content	'0'
	'0'
LRC Check	'1'
	'3'
End 1	(0DH)(CR)
End 0	(0AH)(LF)

Response message (Slave):

Start	'.'
Slave Address	'0'
	'1'
Function	'1'
	'0'
	'0'
	'1'
Starting data address	'1'
	'1'
	'2'
	'0'
Number of data	'0'
	'0'
	'2'
	'D'
LRC Check	'A'
	(0DH)(CR)
End 0	(0AH)(LF)

RTU Mode:

Command message (Master):

Slave Address	01H
Function	10H
Starting data address	01H (High) 12H (Low)
Number of data (In Word)	00H (High) 02H (Low)
Number of data (In Byte)	04H
The first data content	0BH (High) B8H (Low)
The second data content	00H (High) 00H (Low)
CRC Check Low	FCH (Low)
CRC Check High	EBH (High)

Response message (Slave):

Slave Address	01H
Function	10H
Starting data address	01H (High) 12H (Low)
Number of data (In Word)	00H (High) 02H (Low)
CRC Check Low	E0H (Low)
CRC Check High	31H (High)

Please note:

Before and after the transmission in RTU mode, 10ms of silent interval is needed.

■ LRC and CRC transmission Error Checking

The error checking in ASCII communication mode is LRC (Longitudinal Redundancy Check); CRC (Cyclical Redundancy Check) is for RTU communication mode. The algorithm of both is as the following.

LRC (ASCII mode):

Start	'.'
Slave address	'7'
Function	'F'
Starting data address	'0'
	'3'
	'0'
	'5'
Number of data	'C'
	'4'
	'0'
	'0'
LRC Check	'0'
	'1'
	'B'
	'4'
End 1	(0DH)(CR)
End 0	(0AH)(LF)

The LRC algorithm is: add all byte, round down the carry and take 2' s complement. For example, 7FH + 03H + 05H + C4H + 00H + 01H = 14CH, round down carry 1 and take 4CH. 2's complement of 4CH is B4H.

CRC (RTU Mode):

The description of CRC is as the followings:

- Step 1: Load a 16-bits register of FFFFH, which is called **CRC** register.
- Step 2: (The low byte of CRC register) XOR (The first byte of command), and save the result in CRC register.
- Step 3: Right move one bit. Check the least significant bit (LSB) of CRC register. If the bit is 1, then (CRC register) XOR (A001H).
- Step 4: Return to Step 3 until Step 3 has been executed for 8 times. Go to Step 5.
- Step 5: Repeat the procedure from Step 2 to Step 4 until all byte is processing. Get the result of CRC value.

Description: After calculating CRC value, fill in the low word of CRC first in command message, and then fill in the high word of CRC. For example, if the result of CRC algorithm is 3794H, fill in 94H in low word and then 37H in high word.

ADR	01H
CMD	03H
Starting data address	01H (High)
	01H (Low)
Number of data (In Word)	00H (High)
	02H (Low)
CRC Check Low	94H (Low)
CRC Check High	37H (High)

Example of CRC program:

Produce CRC in C language. This function needs two parameters:

unsigned char* data;

unsigned char length

The function returns the CRC value as a type of unsigned integer.

```

unsigned int crc_chk(unsigned char* data, unsigned char length) {
    int j;
    unsigned int reg_crc=0xFFFF;

    while( length-- ) {
        reg_crc^= *data++;
        for (j=0; j<8; j++) {
            if( reg_crc & 0x01 ) { /*LSB(bit 0 ) = 1 */
                reg_crc = (reg_crc >> 1)^0xA001;
            } else {
                reg_crc = (reg_crc>>1);
            }
        }
        return reg_crc;
    }
}

```

PC communication program example:

```

#include<stdio.h>
#include<dos.h>
#include<conio.h>
#include<process.h>
#define PORT 0x03F8      /* the address of COM 1 */
#define THR 0x0000
#define RDR 0x0000
#define BRDL 0x0000
#define IER 0x0001
#define BRDH 0x0001
#define LCR 0x0003
#define MCR 0x0004
#define LSR 0x0005
#define MSR 0x0006
unsigned char rdat[60];
/* read 2 data from address 0200H of ASD with address 1 */
unsigned char tdat[60]={':','0','1','0','3','0','2','0','0','0','0','2','F','8','\r','\n'};

```